Milestone 3 | Detection results

IE 7615 - Group 1 | Evaluation and report: Yosephine Tong

We tested Aditi's saved YOLOv8n model on the held-out Milestone 2 dataset: 36 images containing 113 labelled faces.

Metric	Baseline	Selected settings
mAP@0.5	96.2%	96.2%
mAP@0.5:0.95	86.2%	86.2%
Precision	89.8%	92.9%
Recall	93.8%	92.0%
Mean IoU of correct matches	0.931	0.932

Confidence cutoffs: baseline 0.25; selected 0.50.

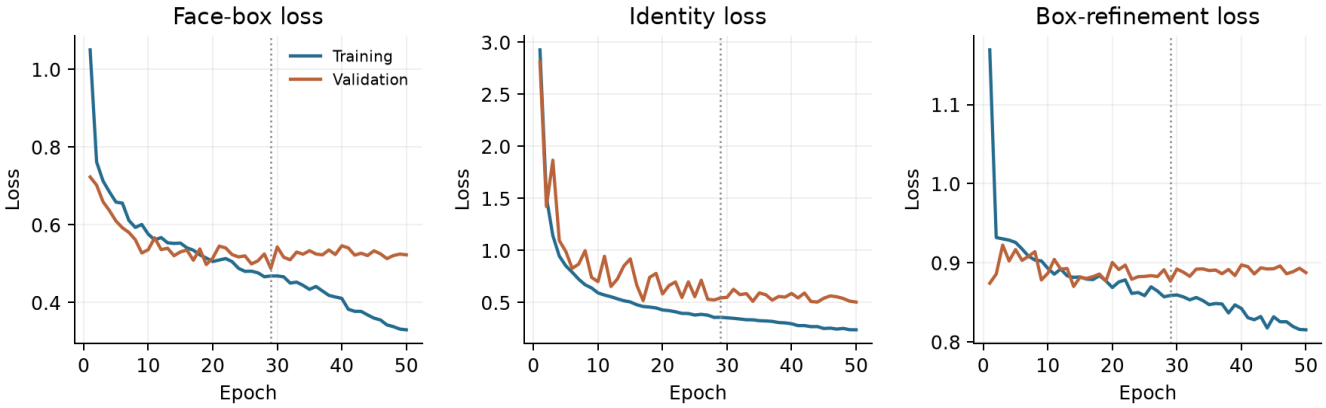
A correct match needs the right identity and box overlap (IoU) of at least 0.50. Mean IoU measures only correct matches, excluding misses and false alarms. mAP uses the full confidence curve, so changing the reporting cutoff alone does not change it.

The selected settings produced fewer false alarms, with a small increase in missed faces. The model weights stayed the same; only the prediction settings were compared.

These results apply to synthetic scenes made from a small set of repeated photos. The labels were generated with a face detector. Strong scores here do not guarantee the same performance on real group photos.

Training and validation loss

The model trained for 50 epochs. Epoch 29 had the best validation score and supplied the checkpoint tested here. The test results were not used to choose it.



Training loss fell, but validation loss stopped improving as much. The dotted line marks the selected checkpoint.

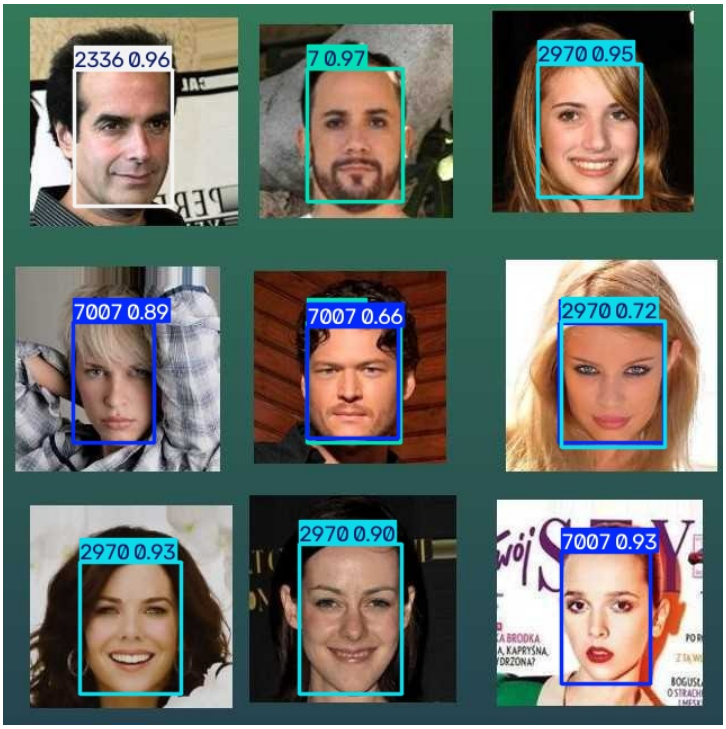
Hyperparameter comparison

Dario's code compared 24 prediction settings on validation, without retraining. The choice balanced precision and recall using F1. Only the baseline and selected settings were then checked on test.

Setting	Compared	Selected
Confidence cutoff	0.10, 0.25, 0.50, 0.75	0.50
Duplicate-box overlap (NMS IoU)	0.30, 0.50, 0.70	0.70
Duplicate filtering	Within each ID / across IDs	Within each ID

TA image check

Using the selected settings, the model placed boxes around all nine faces in the TA image. However, the middle-centre and middle-right faces each received overlapping predictions with different IDs.



No reference identity labels or boxes were supplied, so this is a visual check rather than a scored test. The model can output only its trained identities; a confident prediction can still be wrong.

Detailed scores and run settings are saved in project_milestone03/results/.

Metric reference: docs.ultralytics.com/modes/val